

Motivation

Sequential data is challenging due to its complex patterns. While state-of-the-art classical models achieve high performance, they often require many parameters and significant computational resources. Temporal convolution networks provide an efficient solution by capturing long-range temporal dependencies.

Can quantum computing make TCNs **more efficient** ?

Research Gaps

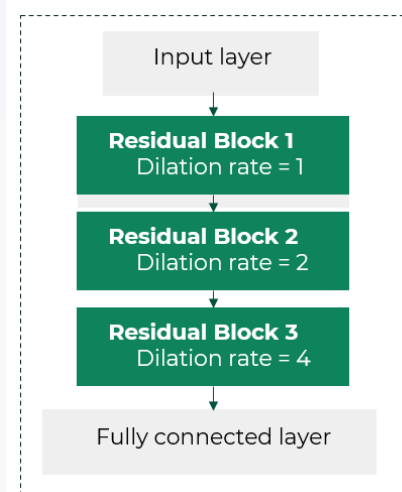
Several studies introduce Quantum TCN architectures, but a fair comparison with classical TCN is **still lacking**.

Paper	Proposed Quantum model	Compared against
Ahmad and Jas, 2025	QTCN	QCNN
Zhuang et al., 2025	Bert-QTCN	Bert-TCN
Yang et al., 2022	QTCN-LSTM	TCN-LSTM

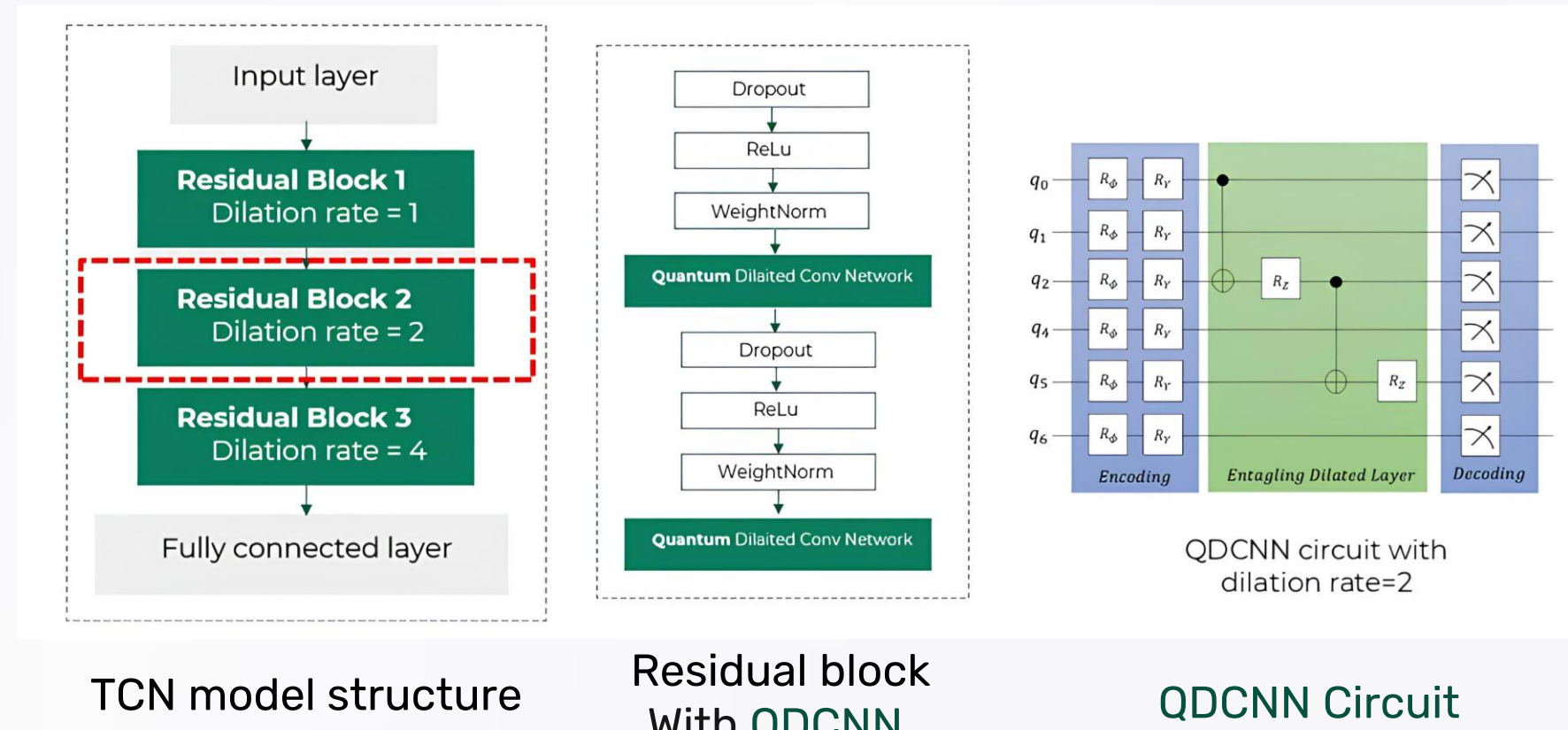
"Better than classical?" Bowles et al., 2024 remains an open question

Where is the quantum part?

Following the hybrid quantum-classical paradigm [Broughton et al., 2021] we replace the core dilated convolution layer with its quantum counterpart (QDCNN) [Chen, 2021].



Quantum Temporal Convolution Network



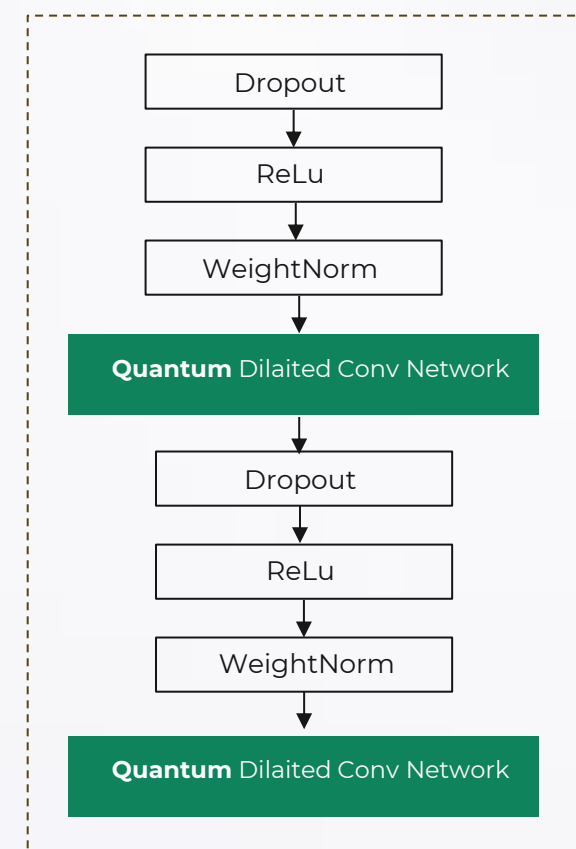
Results and discussion

Experiments were conducted using PennyLane's default.qubit simulator and PyTorch under identical training settings.

Task	LSTM	GRU	RNN	R-Transformer	TCN	QTCN
Adding Problem (loss)	0.164	5.3e-5	0.177	-	5.8e-5	0.139
Sequential MNIST (accuracy)	87.2	96.2	21.5	99.1	99.0	50
JSB Chorales (loss)	8.45	8.43	8.91	-	8.10	0.0026
Nottingham (loss)	3.29	3.46	4.05	2.37	3.07	3.9e-5

On low-complexity tasks, QTCN was less competitive, a trend also reported in prior quantum benchmarking studies [Bowles et al., 2024] . However, it outperformed classical models on more complex and structured datasets.

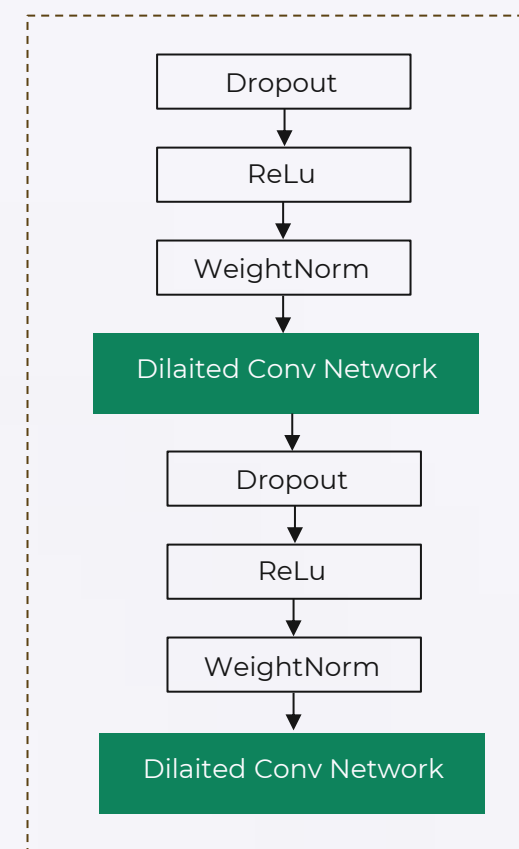
Quantum TCN



Quantum architecture **mirrors** the classical TCN

Ready for fair benchmarking

Classical TCN



Quantum Vs Classical TCN

Task	Classical TCN Parameters	QTCN Parameters
Adding Problem	~70k	~38k
Sequential MNIST	~70k	~38k
JSB Chorales	~300K	~38k
Nottingham	~1M	~38k

QTCN achieves competitive performance with fewer parameters, compared to the classical TCNs. It effectively models complex structured sequences, such as music, with potential for language applications. Its main limitation is the dilation range constrained by the number of qubits.

Benchmarking Datasets

Task	Input	Task
Adding Problem	Sequence of length n depth 2	Compute the sum of the two marked values
Sequential MNIST	784×1 pixel sequence	Classify the digit from the pixel sequence
JSB Chorales	88-bit piano vectors	Predict next notes in polyphonic music
Nottingham	88-bit piano vectors	Predict next notes in folk tunes

Low
↓
High
Complexity

Conclusion

To ensure fair benchmarking, we implemented a mirrored classical TCN architecture for comparison. Results show that QTCN can provide advantages for complex and structured sequential data. By leveraging entanglement as its core mechanism, QTCN offers greater interpretability compared to black-box models, enabling further optimization and future improvements.